Identification of Cryptic Pockets in RAC1 and RHOGAP/GEFs binding interfaces

Elvis A. F. Martis and Stéphane Téletchéa*

US2B, Nantes Université, CNRS, UMR 6286, F-44000 Nantes, France

E-mail: stephane.teletchea@univ-nantes.fr

Abstract

The work described here attempts to locate cryptic pockets in human RAC1 and RAC1-GEF interfaces to enhance the scope of therapeutic interventions in RAC1-associated disease pathways. RAC1 (Ras-related C3 botulinum toxin substrate 1) acts as a molecular switch catalysing by hydrolysing GTP to GDP Owing to the slow nature of GTP hydrolysis and even slower dissociation of GDP, RAC1 is always assisted by Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs). The GEF and GAP binding to RAC1 control the switch mechanism, providing an opportunity to modulate the RAC1 activity in several diseases in which it has been implicated. Since targeting the GTP pocket is challenging due to the large concentration of GTP to compete with the drug molecules, we embarked on finding alternative pockets on either RAC1 alone or in the interface RAC1-GEFs. We found two putative cryptic pockets that are amenable to structure-based drug design. The first cavity is found on RAC1 alone. However, the pocket dynamics were by GEF binding. The second cavity was formed by the RAC1-GEFs interface that provides more GEF-dependent modulation by protein-protein interface binders.

Introduction

- importance of GTPases, their switch mechanism and their interactions with GEFs, GAPs and GDIs

  Small GTPAses act as molecular switches cycling between OFF and ON states. This switching mechnism thus controls a vast variety of processes that control cell growth and differentiation to vesicular and nuclear transport.

- structural biology information about RAC1 and protein-protein interactions with GEFs, GAPs and GDIs

- Therapeutic interventions on RAC1 and their disadvantages
- Binding site information and changes in the binding sites, if any are reported
- Are there any reports on cryptic pockets for RAC1

Methods

Table 1: Detailed system studies

Initial PDB	Protein	Modulator	Size (x, y, z, nb atoms [1])
1HE1[**gappdb**]	RAC1 (ACTIVE)	–	33×34×22, 29000
4YON[**prex1pdb**]	RAC1:PREX1	–	33×34×22, 29000
1FOE[**tiam1pdb**]	RAC1:TIAM1	–	33×34×22, 29000
2VRW[**vav1pdb**]	RAC1:VAV1	–	33×34×22, 29000
4YON (docking)	RAC1:PREX1	NSC23766	33×34×22, 29000
4YON (docking)	RAC1:PREX1	IA-116	33×34×22, 29000
4YON (docking)	RAC1:PREX1	ZINC69391	33×34×22, 29000
4YON (docking)	RAC1:PREX1	EHT1864	33×34×22, 29000

- List of PDBS used. All systems modelled are described in Table **??**.

Preparation of systems for molecular dynamics simulation:

The coordinates for the protein-ligand complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in Table S1 in the supplementary information. The X-ray structures were imported and processed in the Maestro-GUI (Schrödinger suite, 2024). All systems were prepared for MD simulations with the Leap module in AMBERtools23. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field[**ff14sb**].

Generating AMBER-compatible parameters for ligands:

Molecular dynamics simulations:

All complexes were solvated with TIP3P[**tip3p**] waters enclosed in a box with dimensions extending 10 Å beyond any protein atom, and an appropriate number of neutralizing ions were added to maintain the system's neutrality. The complexes were relaxed using three cycles of minimization of 50000 steps each. The first 5000 steps of minimization adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0 Å for the minimization phase. The first cycle of minimization was performed with 25 $kcal/(mol\mathring{A}^2)$ restraining force on the solute atoms, this force constant was reduced to 5 $kcal/(mol\mathring{A}^2)$ in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (table 2) from 0 K to 300 K for 10 ns with 50 $kcal/(mol\mathring{A}^2)$ restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 $ps^{-}1$. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 $kcal/(mol\AA^2)$. In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method[**pme**]. The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled PMEMD code[**pmemd01**, **pmemd02**, **pmemd03**] in AMBER22[**amber22**, **amber**]. All simulations were performed with the SHAKE algorithm[**shake**] to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton's equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Detailed stage of Heating using NVT Ensemble

Stage	Temperature range (K)	Time (ns)
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
stage 4	300 to 300	7.5 to 10

Analysis of MD trajectories:

Analysis of the MD trajectory was performed with the cpptraj[**cpptraj**] module in AMBERTools23. Before the analysis, all solvent atoms and neutralizing ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted Cα root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The convergence of MD trajectories was estimated using the root-mean-squared average correlation (RAC) with mass-weighted RMS averaging of Cα atoms, with the first frame of the trajectory as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations.

Cryptic Pocket Identification:

POVME-2.2.2 [**durrant2011povme**, **durrant2014povme**]

- fragment placement and trying to keep the pocket open to target drug-like molecules
  - Type of pocket and the type of fragment more suitable to keep it open
- Analysis of other MD observations to support pocket volume changes
  - RMSDs for switch regions 1 and 2
  - (perhaps) PCA analysis to explain the switch region dynamics and the pocket volumes

Results and discussion

We studied how different RAC1 experimental structures were comparable to each other by computing their RMSD. This allowed to select X structes in the apo form of Rac1, then Y structures with Rac1 bound to one partner, Z structures where Rac1 was bound to a ligand, and S structures where RAc1 was both bound to a partner protein and a ligand.

- Try to look for a connection/relation between the following
  - switch region dynamics with different binding partners
  - switch region dynamics with pocket volumes

– If we find different pockets that might open and close with different binders

Discussion

Proteins are dynamics by nature, even if RAs members are closely related, their subtle sequence differences are linked to specific

Figures/image1.png

Figure 1: Systems prepared and studied in this article

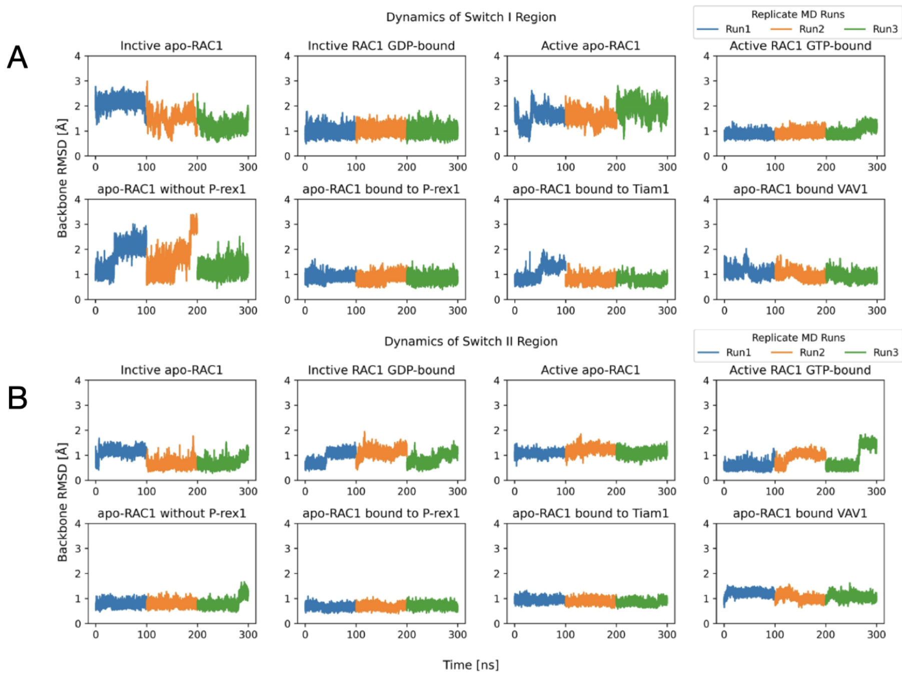

Figure 2: Switch Region Dynamics is RAC1. A) Switch 1 region and B) Switch 2 region

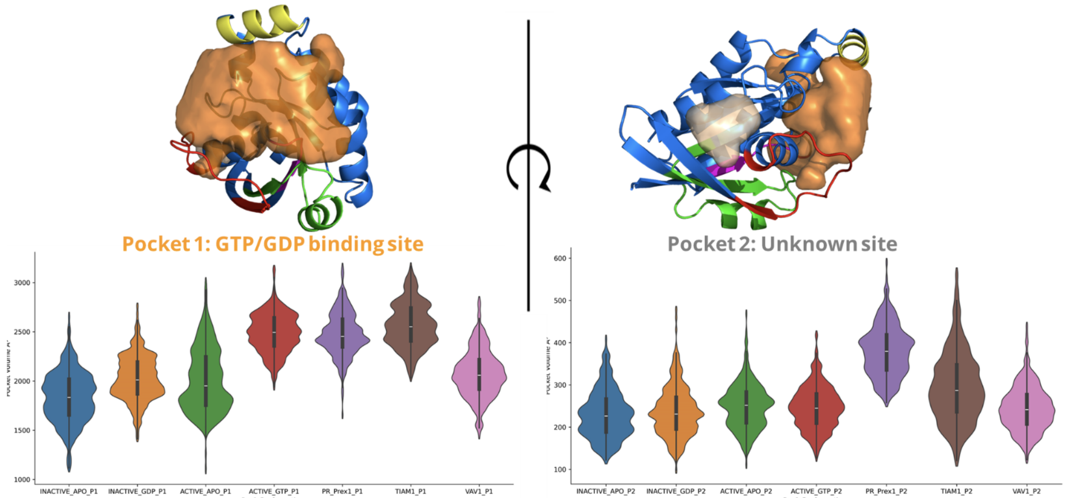

Figure 3: Variation in Pocket Volumes in RAC1 with or without GEF binding partners.

Figures/image1.png

Figure 4: Binding energy of selected ligands in their pockets, and detailed energetics contributions using MMGBSA.

Figures/pocketAnalysis.png

Figure 5: Amino acids involved in pocket definition, and importance of their role in ligand binding. AKA ligand footprint in the pocket of interest.